



MECH MAINT Department
SIMHADRI



Ref: SMTTP/MECH MAINT/2022-23/LMI/BMD/674289

Date:

13/12/2022(17:17:39)

Subject: Revised LMI on Operation and Maintenance Guidelines for Variable Pitch Axial Fans

The LMI on "Operation and Maintenance Guidelines for Variable Pitch Axial Fans " LMI/OPS/MECH/051 has been revised as per revised OGN "COS-ISO-00-OGN/OPS/MECH/051" Rev. No.: 1 Oct. 2022. Revised LMI and OGN is attached.

Put up for kind approval please.

Submitted by:

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All applicable revisions incorporated as per revised OGN.

May please be approved.

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The new LMI has been revised as per revised OGN.

The revisions made in LMI have been mentioned in attached document (LMI on O&M of Variable Pitch axial fans Rev-1 Dec'22) at Page no. 19 (Revision Index).

Submitted for approval please

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In the revised LMI, the last sub-clause under 4.1 is 4.1.12, while in its revision index (it should be read in as page no. 21 and not page-19) the same is mentioned as 4.1.13. Please confirm the correct sub clause no. and make necessary correction in the revised LMI. Also page no '19' is repeating three times at the fag end sheets of the revised LMI. It should read Page no.19, 20 & 21. Please correct the same in the revision enclosed in the Annexures. Rest is OK. Kindly resubmit the revised LMI, after necessary corrections.

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Please do the necessary corrections as per Pre-Page

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LMI revised and attached after necessary corrections.

May please be approved.

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The necessary corrections done i.e. Clause no changed to 4.12 instead of 4.13 in revision index and Page numbering 19-21 corrected.

Submitted for approval.

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Revised LMI may please be accorded approval.

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LMI/OPS/MECH/051

Revision -01 , Dec' 2022

LOCATION MANAGEMENT INSTRUCTION

Operation and Maintenance Guidelines for Variable Pitch Axial Fans

NTPC LIMITED

SIMHADRI SUPER THERMAL POWER PROJECT

NTPC LIMITED

SIMHADRI SUPER THERMAL POWER PROJECT

TECHNICAL COMPLIANCE SYSTEM

LOCATION MANAGEMENT INSTRUCTION - LMI/OGN/OPS/MECH/051

Rev.No.:1 Date: Dec' 2022

***Operation and Maintenance
Guidelines for Variable Pitch
Axial Fans***

Approved for
Implementation by -----
HOP (SSTPP)

Date: -----

Enquiries to : GM(Operation & Maintenance)

LMI APPROVAL DOCUMENT

1	TITLE OF LMI	OPERATION AND MAINTENANCE GUIDE LINES FOR VARIABLE PITCH AXIAL FANS.
2	DOCUMENT NUMBER REVISION/ ISSUE NO & DATE	LMI/OPS/MECH/051 Revision -1, Dec' 2022
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4	PREPARED BY	SIGNATURE: Krishan Lal, Sr. Mgr(BMD)
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6	RECHECKED BY	SIGNATURE: Om Prakash, AGM (BMD)
7	REVIEWED BY	SIGNATURE: Mathew Eipe Kovoov, GM (Maint)
	REVIEWED BY	SIGNATURE: Sanjay Kumar Sinha, GM (O&M)
8	APPROVED FOR ISSUE	SIGNATURE: Girish Chandra Choukse, HOP (SSTPP)
9	DATE OF ISSUE	
10	INDEX LIST	ENCLOSED
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OPERATION AND MAINTENANCE GUIDELINES FOR VARIABLE PITCH AXIAL FANS

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Operation and Maintenance Guidelines for Variable Pitch Axial Fans

1.0 INTRODUCTION

This document aims at providing adequate guidelines for safe operation and maintenance practices for Variable Pitch Axial Flow Fans. Axial fans require periodic inspection and maintenance to ensure their mechanical and aerodynamic integrity for reliable & efficient operation as well as safety precautions. Periodic testing of lube oil, handling of clean air & vibrations within limit are essential for longer life & trouble-free operation of fan. The OEM maintenance routine should be strictly followed for each fan in order to track reoccurring damage mechanisms as well as to prevent them. The maintenance areas include shaft bearings, blade bearings, and the hydraulic blade adjustment device / servomotor and blade replacement.

NTPC Simhadri having **M/S KKK (BHEL Supplied)** make Axial Fans for FD fans and PA fans

2.0 SUPERSEDED DOCUMENT

LMI/OPS/MECH/051 Revision -0, Oct' 2022

3.0 SAFETYASPECT

- 3.1** All personnel that operate, inspect, or maintain the fans should be trained in the plant safety procedures.
- 3.2** Load testing of lifting tools & tackles, supporting structures by Safety / FQA / third party & readiness before usage during maintenance work.
- 3.3** The fan parts should be lifted in the hoisting lugs intended for that purpose.
- 3.4** Proper illumination at workplace is important to ensure safe working.
- 3.5** Permit to work has to be obtained before start of the work by competent person. The permit must be signed by both SAP and CP and a copy and the lock out key has to be kept at worksite.

- 3.6** The main motor and fan auxiliaries must be electrical isolated & locked and the same must be ensured before starting any work.
- 3.7** The respective discharge damper must also be in closed condition & electrically isolated, if working while other fan is in running condition.
- 3.8** Even when a fan is locked out electrically, it may be subject to “wind milling” because of pressure differential across the fan. Therefore, the impeller should be secured to physically restrict rotational movement.
- 3.9** Peptalk of worker before starting the maintenance, discuss all the activities to be carried out/ covered in job safety analysis (JSA) & discuss the risks associated with them.
- 3.10** Inspection doors must not be removed until the rotor is stationary. Before opening the manhole door, ensure fan duct/ housing is not in pressurized condition because of any reason.
- 3.11** Depressurize hoses/pipes containing high-pressure fluids and allow them to cool down to a suitable temperature.
- 3.12** Do not climb the fan and auxiliary equipment without proper safety equipment e.g. scaffolding or safety belts.
- 3.13** Mandatory use of safety belts/ harness with lifeline by workers working at fan casing/height.
- 3.14** Immediately clean the lube oil spills properly during the work to avoid injury due to slippage.
- 3.15** Do not step on the consoles or supports protruding from the surface.
- 3.16** All coupling guards must be fitted in position before cancellation of PTW.
- 3.17** Before cancellation of PTW, it should be ensured that all personnel, scrap and tools must be removed from the fan.
- 3.18** Vibration and performance testing require the fans to be running during the tests. Special precautions must be observed while performing these tests.
- 3.19** Loose cloths that may be sucked/ rolled on the fan rotating parts, must be avoided.
- 3.20** The healthiness of earthing system of motor must be ensured with help of electrical department for safety of man & machine.
- 3.21** The use of 24-volt supply hand lamps shall be ensured. If available use DC

lights for working inside the internal chamber of fan during breakdown/short shutdown.

- 3.22** All spare rotor assemblies of axial fans should be stored indoors.

4.0 MAINTENANCE PRACTICES

The following maintenance practices are to be adopted for variable pitch axial flow fans. Overhauling protocols/critical dimension are to be recorded /maintained as per OEM recommendations. Obtain & Maintain the records/protocols of original dimensions during erection/commissioning stage in newly commissioned fans for reference. Information on WhatsApp to be given as C1 information. **The C-2 form for Rotary Machines must be filled & sent to CC-OS after any breakdown, which caused generation loss or critical component failure. The history of fan maintenance must be maintained & should be kept on BMD dashboard at plant web page.**

The health matrix (as given in annexure) of fans must be filled quarterly & sent to CC-OS.

- 4.1 Actions to be taken prior to fan maintenance** Record the important parameters of fan like vibration level on all bearings of fan & motor, bearing temperature, motor amperage, blade position, lube oil tank temperature, Lube oil pumps discharge pressure, etc before stopping the Fan. Fan bearing vibration should be limited to less than 2.3 mm/sec for new fans being commissioned and after overhaul/repair. Above 4.5 mm/sec vibrations, signature analysis to be done to find out root cause and suitable corrective action to be taken.
- 4.1.2** It is required to record every activity & measurements during maintenance & follow the procedure as given by OEM to avoid any future problems.
- 4.1.3** Check the recent wear debris analysis report.
- 4.1.4** After taking permit to work, the Fan motor Coupling alignment readings may be recorded/ ensured for reference during re-commissioning.
- 4.1.5** Ensure availability of spares, consumables, special tools, welding machine, gas cutting set, service air point & readiness of hoist.
- 4.1.6** In new units, Equipment being critical / precision in nature, Technical assistant services of the OEM should be explored at least for the maintenance / OH of rotary equipment for reliability / availability of equipment and also to familiarize fan maintenance personnel with the new design, best practices, usage of T&P and spare parts planning. **The technical advisor (TA)/ OEM/OES Expert visit report consist of dismantling / assembling procedure & critical clearances etc, is to be prepared & to be**
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Shared with Head CC-OS Boiler. This report shall be circulated among other NTPC stations for knowledge sharing.

4.1.7 OEM/OES supplied or reliable oil seals of Viton Rubber / PU (Polyurethane) to be used in critical equipment's like Fan.

4.1.8 Fan maintenance criteria:

Time Period	Observation	Bearing Replacement	Servomotor servicing
02 Years (Approx. 17500 running Hours)	Break down experienced within 02 years of overhaul then replace respective component. Do RCA with OEM	Yes	Servo motor/ HAD/ Hydraulic Cylinder replacement
4 Years (Approx. 35000 running Hours)	If Fan vibration $\geq 4.5\text{mm/sec}$ (RMS) or axial float is more than recommended value with increasing vibration trend but No break down history.	Yes	Oil seals / O-rings replacement
6 Years (Approx. 52000 running Hours)	All Parameter Normal and no break down history.	Yes	Complete Servicing

Note: As per BS ISO 10816-3, the fan bearing vibrations should be limited to less than 2.3 mm/sec for new fans being commissioned and after overhaul/repair or OEM recommended, whichever is less. Signature analysis to be done to find out root cause of vibrations. If vibrations are $\geq 4.5\text{mm/s}$ suitable corrective action to be taken. If required OEM Engineer to be called for necessary guidance & supervision.

Servicing of blade bearing assembly is to be carried out with servicing of main bearing assembly

4.1.9 The following activities must be carried out in every fan overhaul-

- Coupling inspection & alignment
- Inspection of oil leakages from bearing housing simmer rings/oil seals / drain line bellow
- Inspection of oil leakages from blade bearing seals (before and after maintenance by operating LOP/Blade pitch control system)
- Lube oil system servicing & /tank cleaning
- Blade cleaning & DP Testing after dismantling

- Inspection of servomotor/HAD linkage system
 - Silencer inspection & cleaning
 - Cleaning and repair of protective canopy over silencer
 - SCAPH cleaning & testing
 - Other activities as per requirement / OEM recommendations
- 4.1.10 Fan dismantling & assemble work must be carried out at covered place free from dust/ash.
- 4.1.11 Ensure direct earthing from welding machine, while carrying welding at any fan part.
- 4.1.12 Spare fan assemblies & subassemblies like bearing housing, impeller hub, servomotor must be stored & preserved as per OEM recommendations.

4.2 Fan inspection before & after maintenance

- 4.2.1 Inspect any oil leakage from bearing housing / servomotor seals/ blade spindles of rotor assembly.
- 4.2.2 Checking of Impellers and blades for deposits and rust.
- 4.2.3 Marking the assembled components by match marks before dismantling.
- 4.2.4 Cleaning of fan after maintenance for foreign materials like loose bolts, nuts, welding electrodes, any debris, etc.
- 4.2.5 Inspection of cleanliness should be done in blade full open condition & by slowly rotating the impeller. Special care should be taken while checking double stage fans.
- 4.2.6 Ensure fan suction side nut bolts are properly tightened with washers to prevent loosening during fan operation. All loose materials/structures, projections, if any, at suction chamber must be removed and cleared.
- 4.2.7 Cleaning of fan discharge ducts after completion of work in Air Preheaters during unit overhaul should be done to avoid entrapment of foreign material.
- 4.2.8 Draining facility should be ensured to prevent accumulation of water in fan casings & duct during water washing of Air Preheaters.
- 4.2.9 Protections & interlocks of fans must be tested after maintenance work.
- 4.2.10 During re-commissioning of the units after maintenance, coupled trial run of fans should be taken before boiler light-up in presence of Maintenance and Operation personnel. Check the position of oil/ water line valves & ensure required oil/ water flowing in return lines. Line up of duct system to be ensured before trial run.
- 4.2.11 The cleaning of SCAPHs should be ensured, which may affect fan performance.
- 4.2.12 The foundation spring isolator units do not require any special maintenance. Routine Inspection of the surface should be carried out for corrosion /painting/ damages. Spring units should be cleaned by air (Not by water). A record of spring height wrt original settings during erection/commissioning stage/as per drawing may be recorded/verified during overhauling, if available. In case of any major load (Machine Parts) to be removed from foundation, the pre-stressing bolts shall be hand tightened and later should be released after keeping the load on foundation.

4.3 Disassembly and Assembly of the Blades

- 4.3.1 Ensure correct marking/serial numbers on blade before removal
- 4.3.2 Cleaning & visual inspection of blades for erosion & corrosion
- 4.3.3 DP Testing at root & full length of blades
- 4.3.4 In case of cracks/pitting/ex-foliation/erosion, replace complete set of blades, if required.
- 4.3.5 Before assembling the blades, keep the servomotor in full close position.
- 4.3.6 Fixing of blades as per serial number & record of root/ blade base circumference gap as per instructions of OEM.
- 4.3.7 Screws must be coated with anti-seize compound and have to be tightened to the specified torque with torque wrench. Make sure that the screws are loaded uniformly. Blades screw must be replaced, while replacing the blades.
- 4.3.8 Check the blades for excessive looseness radially & maintain as per drawing from blade shaft locknut
- 4.3.9 Ensure that the blades are positioned and tightened in identical position to avoid uneven gap between blades. In case of double stage fans blades, gap must be same for both the stages.
- 4.3.10 After fixing all the blades, tip clearance to be measured for all the blades as per instructions of OEM. Blade clearance of longest blade should be taken at different casing points.
- 4.3.11 Blade pitch mechanism should not operate, when hydraulic oil system is not in service.
- 4.3.12 The actual blade positions must be calibrated with respect to control room reading and must be identical in both fans. Blade pitch actuator & feedback mechanism coupling (if provided) must be checked by C&I in every OH.
- 4.3.13 In axial type FD & PA fan, Aluminum alloy blades should be used after anodizing & painting to prevent corrosion, if experiencing corrosion in consultation with COS Boiler. If axial type ID & FGD Booster fan are having dust erosion on blades, erosion prevention coating to increase life to be done in consultation with COS Boiler.

4.4 Servicing of Lube oil System & Control Oil system

- 4.4.1 Thorough cleaning of lube oil tank should be carried out by good quality cloth. Never clean the tank with the help of cotton waste to prevent frequent filter chocking problem. Any rusting of tank to be removed and coated with rust preventive painting.
- 4.4.2 Attend all lube oil leakages & other defects based on the requirements.
- 4.4.3 All the view glasses in lube oil system should also be cleaned.
- 4.4.4 Inspection, hydraulic testing & replacement of O-rings /gaskets of lube oil coolers after cleaning of tube bundle.
- 4.4.5 The breather /cap of lube tank must be cleaned & in position.
- 4.4.6 Filter/ strainer cleaning, inspection & replacement based on condition. If there are paper filters in the system then these must be replaced, never try to clean the paper filters. Also maintain sufficient stock of the paper filters.
- 4.4.7 Ensure correct grade of lube oil in case of new oil & its moisture content before use. A permanent board (lubrication chart) should be displayed in lube oil storage area at site, showing use of lube oil/ grease grade equipment/subassembly wise.
- 4.4.8 Never fill the oil more than maximum level.
- 4.4.9 Monthly oil monitoring for moisture & mechanical impurities (MI)/ NAS and Quarterly oil monitoring through wear debris analysis must be followed strictly. The MI/ NAS should be less than 100 PPM/ 7 NAS. The safe limit of PQ index for fans is 25 (As given in Netra WDA report). The moisture content should be less than 100 PPM or as recommended by OEM. Oil should be clear and transparent and should not be hazy. Centrifuging may be done to remove moisture. In case of any abnormality in the WDA, necessary corrective action has to be taken. Oil sample should be collected from system before filters.
- 4.4.10 If hydraulic oil system is given separately, then same procedure should be adopted as given above. Only monthly oil monitoring for moisture & MI is required.
- 4.4.11 Use correct pressure rating & length of oil hoses for hydraulic & lube oil systems. Also prevent rubbing of hoses & other lube oil lines/ pipes with each other or other surfaces by providing permanent soft clamping. Ensure fixing of protective sheath of proper grade in case rubbing can't be avoided. Replace all the lube / control oil hoses with replacement of scheduled fan assembly.
- 4.4.12 Ensure servicing & calibration of all the pressure gauges by C&I.
- 4.4.13 During re-commissioning of lube oil system, one must ensure that oil is coming back to tank from all return oil lines.

4.5 Silencer / Suction Hood Inspection & Cleaning

- 4.5.1 Scaffolding should be erected for proper inspection of silencer.
- 4.5.2 Thorough inspection of silencer supports & Repair if required.
- 4.5.3 Cleaning of silencer suction manifold by compressed air.
- 4.5.4 Inspection & cleaning of canopy / roof of suction duct and repair if any damages found or roof/canopy sheet is found bent in concave shape.
- 4.5.5 No water, ash or any material should be accumulated on the top of silencer /canopy roof, which may lead to damage of blades due to damage of roof/canopy. **Also, an additional highly inclined plate (55 degree from horizontal) should be fabricated at the top of the existing canopy, which shall be supported on beam, especially for the fans located below the TPs & CHP galleries and fall of foreign material is expected.**
- 4.5.6 In newly installed fan, check left out foreign material in the expansion joints and temporary angles/ supports welded in the silencer/ transition ducts etc should be removed.

4.6 Maintenance of Rotor Assembly

- 4.6.1 Casing of the fan housing should be lifted carefully within balanced & vertical condition. Hoists installed for the purpose may be used as far as possible. Casing lifting directly from hydra should be avoided.
- 4.6.2 Care of expansion joints should be taken for freeness & air leakages. (Ensure/check that sufficient flexibility available in expansion joints and also ensure no leakages in the joints).
- 4.6.3 Casing should not touch with blades during removal & re-fixing.
- 4.6.4 The rotor assembly must be removed and suspended securely balanced with safety.
- 4.6.5 Make arrangements to heat the bearing elements with the help of induction heater. Never exceed temperature of 100 deg centigrade. Ensure the temperature and allow sufficient soaking time before mounting.
- 4.6.6 The bearing clearances before & after mounting must be ensured.
- 4.6.7 Both end simmer rings/ O-rings should be replaced as per specific design of Fan.
- 4.6.8 For the hub mounting data, dimensions and permissible pressure, must refer to relevant drawings.
- 4.6.9 Check the bearing housing axial float & should be within prescribed limit.

- 4.6.10 Ensure inspection & cleaning of guide vanes.
 - 4.6.11 DPT of bearing housing mounting flange should be done. The base of Fan part is to be tightened without strain as per torque indicated.
 - 4.6.12 There should be uniform axial gaps between rotor and stator parts.
 - 4.6.13 The outlet lube oil flange connection of housing and rubber bellow / pipeline should be checked. It should be carefully/keenly/thoroughly checked for oil leakages due to congested space. Lub oil drain line expansion bellow must be replaced, while replacing scheduled fan assembly.
 - 4.6.14 Check the verticality of impeller housing / horizontality of shaft during assembly.
 - 4.6.15 After setting of correct coupling gap, match the marks on the coupling parts. Alignment of couplings should be carried out within 0.05 mm or as per OEM guidelines.
 - 4.6.16 Fan lube oil system trial should be taken for checking oil leakages from bearings, hoses, servomotor etc before placement of upper half casing.
 - 4.6.17 Insulation should be provided properly with cladding on the casing/ duct after maintenance. Use beading & sealing compound to prevent water ingress, if required.
 - 4.7 Servicing of servomotor / Hyd Adjustment Device**
 - 4.7.1 Replace all the oil seals and bearings or as per OEM recommendation. There should not be any scratch on piston.
 - 4.7.2 Ensure proper setting of mechanical stoppers as per stroke of hydraulic adjustment device assembly, if provided.
 - 4.7.3 Ensure alignment of servomotor/ hydraulic adjustment device as per procedure recommended in manual.
 - 4.7.4 Ensure there is no pre-load in servomotor oil head assembly introduced by flexible hoses.
 - 4.7.5 If required OEM services may be taken for hydraulic cylinder with rotating oil seal servicing/replacement & alignment, being a highly critical assembly.
 - 4.8 Servicing of blade bearing assembly**
 - 4.8.1 Removal, cleaning & inspection the lever assembly, locknut, counter weight, shaft, connecting ring and distance ring assembly.
 - 4.8.2 Ensure correct grade of lubricants in different parts of lever assembly.
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- 4.8.3 All oil seals and O-rings should be replaced or as per OEM recommendation.
 - 4.8.4 Hydraulic testing of each blade spindle is to be carried out at given pressure as OEM recommendations.
 - 4.8.5 Free movement of all blade spindles and axial float should be ensured during servicing.
 - 4.8.6 In blade pitch linkage mechanism, pin with split pin or bolts and nuts with split pins to be used for additional locking. Any back lash in coupling/ linkage should be attended to avoid fluctuation in blade pitch.
 - 4.9 Fan Internal Inspection during short shutdowns**
 - 4.9.1 Cleaning should be done inside duct, blades and servomotor area.
 - 4.9.2 Inspection of servomotor link rod assembly, rod end bearings etc. Replaced rod end bearings of link rod in every OH.
 - 4.9.3 Inspection for oil leakages from servomotor, hoses, bearing housing etc.
 - 4.9.4 There should not be mechanical backlash in the linkages of hydraulic adjustment device and actuator. Locking pin and bolts of Servomotor operating link rod should be inspected during every available opportunity. Ensure tightness of nut bolts in fan suction side also.
 - 4.10 Precautions during operation**
 - 4.10.1 Approved Engineering limits for alarm, protections & interlocks of fans on high vibrations & high bearing temperature should not be altered.
 - 4.10.2 Never run the fan with high vibrations.
 - 4.10.3 Fan should not be permitted to run in stall condition. Stall probes with adequate purging system for all fans including FD&PA fans to be ensured. Stall warning system with mimic in DCS to be ensured incorporating operating limits & operating point in line with fan performance curves.**
 - 4.10.4 Special care should be taken while one fan is running and another fan is to be started. Even here the sequence of operation must be such that neither of the two fans runs in the unstable range of stalling zone.
 - 4.10.5 Oil leakage from Servomotor of FD/PA Fans results in plugging of SCAPH/APH baskets. Major oil leakages should be immediately identified and attended.
 - 4.10.6 De-linking of blade pitch mechanism in PA and FD Fans should be attended immediately to avoid damage to fan components due to stalling during part-load operation / runback.
 - 4.10.7 If the bearing temperatures of fan are running high, then check the lub oil flow/ pressure & also ensure cooling water is flowing through lub oil cooler.
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4.11 Checks / Inspections during Long shut down/RSD

- 4.11.1 Once in 15 days, run the each lub oil pump for half an hour and operate the hydraulic adjustment device/ servomotor from blades full open to full close of blades.
- 4.11.2 Periodically exercise Fan's by rotating it several times and leaving it a different (90-degree) angle, if shut down is more than 15 days.
- 4.11.3 Testing of lub oil quality (Moisture & MI) by chemistry. Oil sampling for WDA analysis on Quarterly basis to be continued. Oil sample to be taken with luboil pump in service.
- 4.11.4 The outside surfaces of the fan parts should be checked for corrosion. If found, then take anti-corrosion measures.
- 4.11.5 If vibrations of any fan are on higher side, then root cause analysis & action plan to be prepared after vibration analysis., while shutting down the unit.

5.0 DEALING WITH HIGH FAN VIBRATIONS

Vibration is the best indicator of overall mechanical condition and the earliest indicator of the developing defects. First of all, the vibration signature of a machine should be taken & analyse for possible causes of high vibration.

Most possible reasons of vibrations occur in fan units are listed below.

- a) Bearing failures
- b) Loose components
- c) Imbalance
- d) Worn/damaged parts
- e) Misalignment or bent shaft
- f) Improper Lubrication
- g) Improper clearance of moving parts.
- h) Excitation of a resonant frequency

Operation and Maintenance Guidelines for Variable Pitch Axial Fans

- i) Corrosion, erosion, high/low cycle fatigue effects
- j) Flow induced vibrations from rotating stall
- k) Electrical faults in motors

- l) Loose foundation bolts
- m) Other Foundation problems
- n) Material deposition on the rotor/blade

6.0 Special T&P for maintenance of axial fan

Special T&P for maintenance of axial fan as given below should be procured.
Usage of the same must be ensured, wherever it is available.

SI No	Description	Quantity
1	Hydraulic Jack with pump, hoses and connectors of 50/75/100MTcapacity	1 Set
2	Hydraulic Jack with pump, hoses and connectors of 200MTcapacity	1 Set
3	Laser Alignment Kit with pre-cut SS Shims of different sizes	1 Set
4	Infra-red Non-contact Thermometer	1 No
5	Portable Vibration Analyzer	1 No
6	Special Tool Kit for Fans (As provided by OEM)	1 No for each model Fan
7	Hydraulic Nut (Size- depends on fan model)	1 No for each model Fan
8	Bearing Induction Heater	1 Set
9	Hydraulic Pullers	1 Set of each capacity
10	Pneumatic/ mechanical Torque Wrenches	1 Set
11	Oil injection kit	4 Set
12	Bore gauge, Micro meter, Vernier caliper	1 Set
13	Hydraulic nut / Hollow jack as per requirement	1 Set
14	Master level	2 No
15	Trolley mounted Portable Compressor	1 No
16	Portable Oil Filtration Unit	1 No
17	Arrangement (Jig/fixture) for rotation of fan during reserve shut down/long shutdown	01no for each type of fan

7.0 REVIEW

The Head of Project will be reviewing authority for this document on 2-yearly basis or as necessary.

Check List for overhauling of Axial Fans

Check list of activities should be made for overhauling of fans based on O&M manual/ OEM recommendation/experience with critical dimensions. It should cover all the Checking & inspection points, depending upon the model and protocols also to be made. Sample checklist is given in Annexure-I.

AXIAL FAN CHECK LIST

Project : Unit : Fan :

Fan Type : Speed : Period :

1. SHAFT WITH BEARING ASSY. (IN CASE OF REPLACEMENT OF BEARINGS)

a) Bearing Designation & Condition:

DE Bearing : New /Old
NDE Bearing : New /Old

b) Bearing Seating Areas (in mm):

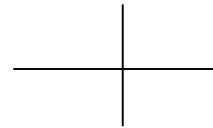
Location	Shaft OD	Bearing ID	Bearing OD	Housing ID
DE Bearing As per Drg				
NDE Bearing As per Drg				
DE Bearing Actual				
NDE Bearing Actual				
Coupling	Shaft OD	Coupling ID		

c) Bearing clearance:

Location	Bearing Make	As per Design	Before mounting	After mounting
DE Bearing				
NDE Bearing				

d) Torque Tightening of Bolts as per drawing :

e) Bearing Face Out :



f) Axial Float

As per drawing :

Actual :

g) DPT OF BEARING HOUSING MOUNTING FLANGE : Done/ Not Done

2. HUB MOUNTING DIMENSION

Location	As per drawing	Actual
DE		
NDE		

3. BLADE BEARING ASSY. (To be checked for each spindle)

a) Tightness of lock nut, lock screw and lever bolts: Done/ Not Done

b) Smooth operation of each blade shaft : Checked/ Not Checked

c) Hyd. test of each blade shaft : Done/ Not Done

4. SERVOMOTOR / Hydraulic cylinder ASSY.

a) Free rotation of control rod : Checked/ Not Checked

b) Torque tightening of bolts : Done/ Not Done

c) Oil leakage : Yes/No

d) Operation of servomotor : Checked/ Not Checked

e) Stroke length: As per Drawing :mm

Actualmm

f) Control oil Flexible hose : Replaced / Not Replaced

g) Inspection of servomotor linkage assy including rod end bearings : Done/ Not done

h) Oil leakage after connecting the LOS to the servomotor and Brg. Assy : Yes/No

i) Servomotor operating time with connecting to the LOSsec

j) Actuator operating time without connecting to servomotor sec

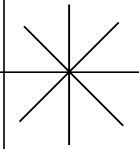
k) Servomotor stroke length after setting mechanical stoppermm

5. BLADE

a) Blade set : Replaced / Not Replaced

b) DPT of full blade : Checked/ Not Checked

c) Blade & Rotor Clearance:

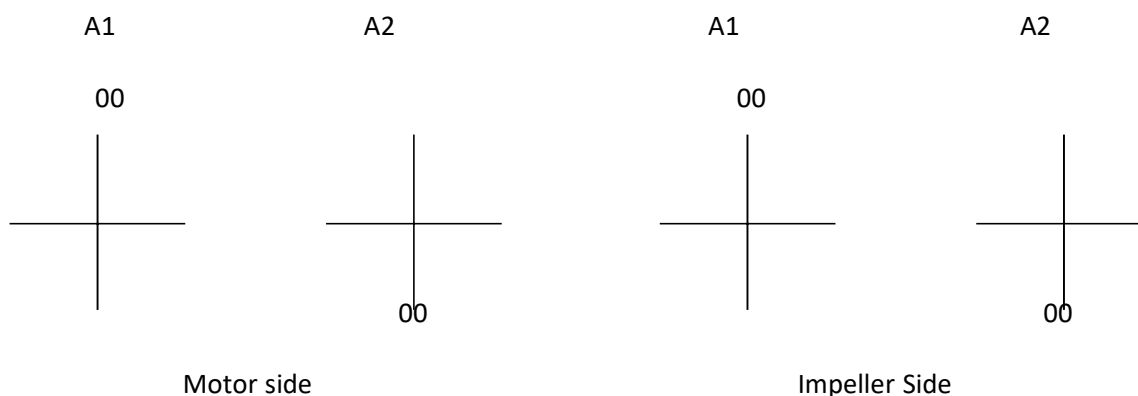
Blade no	Stage 1 Open / Closed	Stage 2 Open / Closed	Root clearance		Blade boss to hub cavity		Rotor to stator clearance		Impeller to Bullet clearance	Impeller to diffuser core clearance	<u>DRIVE END</u> Tip Clear w r t Longest blade no:
			DE	NDE	DE	NDE	DE	NDE			
01											
02											
03											
04											
05											
06											
07											
08											
09											
10											
11											<u>NON DRIVE END</u> Tip Clear w.r.t Longest blade no:
12											
13											
14											
15											
16											
17											
18											
19											
20											
21											
22											
23											

6. COUPLING

Gap as per drg mm

Motor /Impeller side :mm/mm

Coupling Alignment :

**7. LUB OIL/CONTROL OILSYSTEM**

- | | | | |
|----|--|---|-----------------|
| a) | Cleaning of filters | : | Done / Not Done |
| b) | Cleaning of cooler | : | Done / Not Done |
| c) | Cleaning of tank | : | Done / Not Done |
| d) | Hydro test of cooler | : | Done / Not Done |
| e) | Lube oil pipe line cleaning &flushing: | : | Done / Not Done |
| f) | Oil (New /Old centrifuged) | : | New/Old |

8. MISCELLANEOUSWORK

- | | | | |
|----|-------------------------------------|---|-----------------|
| a) | Inspection of foundation bolts | : | Done / Not Done |
| b) | Inspection of all fan duct supports | : | Done / Not Done |
| c) | Cleaning & Inspection of Silencer | : | Done / Not Done |
| d) | Identification mark on the assembly | : | Done / Not Done |
| e) | List of the replaced spares | : | |

11. TRIAL RUN**a) Vibration Readings(Pre-OH):**

Fan Description	Location	MDE	MNDE	Reading at Fan Foot
	Horizontal (D/V)			
	Vertical (D/V)			
	Axial (D/V)			

b) Vibration Readings (Motor Trial):

Fan Description	Location	MDE	MNDE
	Horizontal (D/V)		
	Vertical (D/V)		
	Axial (D/V)		

c) Vibration Readings (Post OH):

Fan Description	Location	MDE	MNDE	Reading At Fan Foot
	Horizontal (D/V)			
	Vertical (D/V)			
	Axial (D/V)			

d) Bearing Temperature:

Fan Description	Location	Temperature
	FDE	
	FNDE	
	MDE	
	MNDE	

NTPC		LMI		SIMHADRI	
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SL No.	Rev. no.	Date	Details	Approved by	
1.	REV 1.00	Dec'22	Section 4.0 Information sharing clause added.		
			Section 4.1.8 Fan maintenance criteria modified		
			Section 4.1.12 Storage & Preservation clause added		
			Section 4.2.10 Recommissioning clause has been changed		
			Section 4.3.7 Blades screw must be replaced, while replacing the blades		
			Section 4.3.13 added for blade coating to enhance the life of blades		
			Section 4.4.9 Value of oil MI/ NAS added		
			Section 4.4.11 Criteria for Replacement of fan oil hoses added.		
			Section 4.6.13 Criteria for Lub oil drain line expansion bellow replacement added		